
16 An Introduction to Groups

16.1 Multiplicative Groups

Recall that $\mathbb{Z}_n$ is the set of integers $\{0, 1, \dots, n-1\}$ with the operations modulo n . In particular the operation $+_n$ takes two elements of $\mathbb{Z}_n$ and produces an element of $\mathbb{Z}_n$. Note that the element 0 is the additive identity (meaning $a +_n 0 = a$), and, if a is not 0, $n - a$ is the additive inverse of a (meaning $a +_n (n - a) = 0$).

We now define $\mathbb{Z}_n^*$. This is the set of integers in the range 1 up to $n - 1$ that are relatively prime to n . Note that if a and b are integers and neither has a common factor with n , then their product still has no common factor with n . This means that if we take a and b in $\mathbb{Z}_n^*$, then their product $a \cdot_n b$, meaning the remainder when ab is divided by n , is itself an element of $\mathbb{Z}_n^*$. So we get a multiplication table, as before. (Note that addition in $\mathbb{Z}_n^*$ is not allowed.)

EXAMPLE 16.1. Here is the table for $\mathbb{Z}_{12}^*$

	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

Note that the fact that 1 appears in every row is to be expected. We argued earlier that in the multiplication table for $\mathbb{Z}_n$ the rows where every element appears is precisely the rows corresponding to elements relatively prime to n .

16.2 Definition of Group

The structures $\mathbb{Z}_n$ and $\mathbb{Z}_n^*$ are examples of a more general situation. A **group** is a set of elements G together with a single binary operation, call it $\star$, such that it obeys the following rules:

1. G is **closed** under the operation $\star$: that is, for all $a, b \in G$ it holds that $a \star b \in G$;
2. **Associative** law: $a \star (b \star c) = (a \star b) \star c$ for all a, b, c ;

3. There is an **identity** e such that for all $a \in G$ we have $a \star e = e \star a = a$; and
4. **Inverses** exist: for all $a \in G$, there is an a^{-1} such that $a \star a^{-1} = a^{-1} \star a = e$.

Note that commutativity is not assumed. Indeed, a group is called **abelian** if it satisfies:

5. Commutative law: $a \star b = b \star a$ for all a, b

In many of our examples, the group operation is addition or multiplication. For example:

- The *positive real numbers* form an abelian group under *multiplication*. Here 1 is the identity, and the inverse of r is $1/r$. Note that all the real numbers does not work, since 0 has no inverse.
- The *integers* form an abelian group under *addition*. Here 0 is the identity, and the inverse of r is $-r$.
- The set of all 2×2 nonsingular *matrices* forms a nonabelian group under matrix *multiplication*. The identity matrix I is the identity, and the inverse of matrix A is A^{-1} .

Notation for the operation: We use the standard symbol if it is a common specific operation, like addition or multiplication. Otherwise, we often use “implicit multiplication” notation: with elements simply written next to each other. The associative law means we can omit brackets completely.

16.3 Basic Facts

Some fundamental properties are the following:

Lemma 16.1

- (a) *The identity element is unique.*
- (b) *The inverse of an element is unique.*
- (c) $(a^{-1})^{-1} = a$.
- (d) $(ab)^{-1} = b^{-1}a^{-1}$.

PROOF. We show (d). The rest are left as an exercise. To show that something is the inverse, one needs to show that multiplying by it produces the identity. So,

$$(ab) \star (b^{-1}a^{-1}) = abb^{-1}a^{-1} = aea^{-1} = e.$$

Similarly, $(b^{-1}a^{-1})(ab) = e$. $\diamond$

Note that because we do not assume commutativity, we have to be explicit about whether we are doing the operation on the left or on the right. Nevertheless, $a \star b = a \star c$ implies, multiplying on the left by a^{-1} , that $b = c$. This is often referred to:

Cancellation Law: if $a \star b = a \star c$ then $b = c$; if $b \star a = c \star a$ then $b = c$.

One can produce a **table**, where the (i, j) entry gives the result of the operation applied to the i^{th} entry and the j^{th} entry. Note that each row contains every element exactly once. Why? Every element must be there, since, for any given a and b , the equation $a \star x = b$ has a solution, namely $x = a^{-1} \star b$. By the cancellation law, this solution is unique.

For example, here is the table for a group with four elements $\{e, a, b, c\}$.

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

16.4 Subgroups

An important concept is a **subgroup**. This is a subset of the elements that contains the identity, and is closed under multiplication and inverses. For example, for the group $\mathbb{Z}$ with addition, the even integers form a subgroup. In general there are always at least two subgroups: the group itself and $\{e\}$. These are called the **trivial** subgroups.

EXAMPLE 16.2. *The subgroups of $\mathbb{Z}_n$*

The subset of $\mathbb{Z}_n$ consisting of all multiples of a is a subgroup for any a .

EXAMPLE 16.3.

The set of 2×2 matrices with determinant 1 form a subgroup of the set of all nonsingular 2×2 matrices.

A group/subgroup is generated by a set X if it consists of all arbitrary products of elements in X and their inverses. A group that is generated by a single element is called a **cyclic group**.

We use the notation a^n as the notation for n a 's multiplied together. Consider the sequence $e, a, a^2, a^3, a^4, \dots$. If the group is finite, there must come a point in the sequence that we

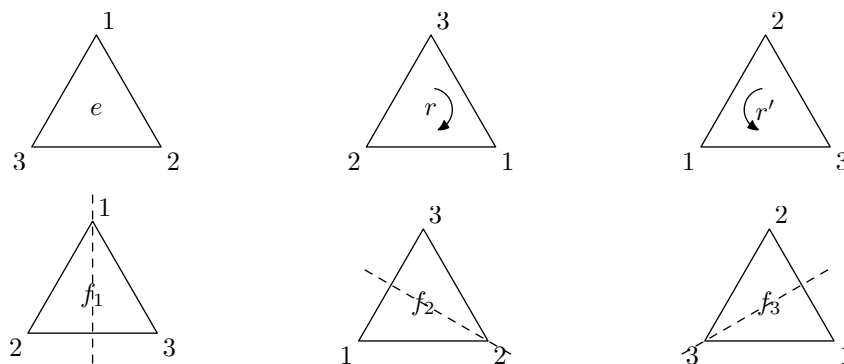
have a repeat. Suppose that first repeat is a^n , and that a^n equals a^m for $m < n$. Then by cancellation $a^{n-m} = e$. So, actually there is no repeat until we reach e . The first positive power of a that equals the identity is called the **order** of a . We also use the term **order** of a group to be the number of elements in the overall group. Thus, the order of an element a is the order of the subgroup generated by a .

16.5 The Dihedral Group

We conclude with an example of a nonabelian group. The **dihedral group** D_n is the set of “symmetries” of a regular n -gon: the motions that leave the n -gon looking unchanged. One can rotate the polygon $360/n$ degrees, or indeed any multiple thereof. One can also think of reflection around various axes. Specifically, the dihedral group has $2n$ elements. There are $n - 1$ **rotations** and n flips. For motions m_1 and m_2 , we define $m_1 \star m_2$ to be the motion obtained by m_1 followed by m_2 .

EXAMPLE 16.4. Consider D_3 .

This is the symmetry group of an equilateral triangle. Say r is the rotation 120 degrees clockwise, r' the rotation 120 degrees counterclockwise, and f_i the flip around an axis through the center and corner labeled i . Then we get the pictures and table.



	e	r	r'	f_1	f_2	f_3
e	e	r	r'	f_1	f_2	f_3
r	r	r'	e	f_2	f_3	f_1
r'	r'	e	r	f_3	f_1	f_2
f_1	f_1	f_3	f_2	e	r'	r
f_2	f_2	f_1	f_3	r	e	r'
f_3	f_3	f_2	f_1	r'	r	e

In general, in D_n a flip is its own inverse. The inverse of a rotation is another rotation. Other group properties can be checked.

Exercises

- 16.1. (a) Show that the identity element is unique.
 (b) Show that the inverse of an element is unique.

16.2. Explain why the following is not a group:

	e	a	b	c
e	e	a	b	c
a	a	a	e	e
b	b	e	b	e
c	c	e	e	c

16.3. Consider the multiplicative group $\mathbb{Z}_{15}^*$

- (a) List the elements of the group.
 (b) Give the identity element.
 (c) Give the inverse of 7.
 (d) Give an element other than the identity that is self-inverse.

16.4. Let $q = 2^{100}$. Determine how many elements of $\mathbb{Z}_q$ have order q .

- 16.5. (a) Show that the intersection of two subgroups is again a subgroup.
 (b) What about union?

16.6. Show that every subgroup of $\mathbb{Z}_n$ is given by the set of multiples of some integer a .

16.7. Show that any subgroup of a cyclic group is itself cyclic.

16.8. Determine the group of symmetries of the following shapes:

(a) The letter: **S**

(b) The plus sign: **+**

(c) A flower 

(d) This shape is infinite: $\cdots \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \cdots$

16.9. Consider the group of 2×2 matrices with nonzero determinant.

- (a) Give an element of order 2.
- (b) Give a nontrivial/proper subgroup.

16.10. Consider the following table for a group.

	e	a	b	c	d	f	g	h
e	e	a	b	c	d	f	g	h
a	a	e	c	b	f	d	h	g
b	b	c	a	e	g	h	f	d
c	c	b	e	a	h	g	d	f
d	d	f	h	g	a	e	b	c
f	f	d	g	h	e	a	c	b
g	g	h	d	f	c	b	a	e
h	h	g	f	d	b	c	e	a

- (a) What is the identity?
 - (b) Give an element of order 4.
 - (c) Is the group abelian? (Justify your answer.)
 - (d) Explain why we know this is not the dihedral group D_4 .
- 16.11. (a) Determine all the subgroups of the dihedral group D_4 .
 (b) Determine all the subgroups of the dihedral group D_5 .